Sophia Mena

sophiacmena@gmail.com | (305) 607 - 4811 | https://sophiamena04.github.io

Education

Florida State University Tallahassee, FL
Bachelor of Science in Applied Mathematics, Minors in Spanish and Physics May 2026

Relevant Coursework: Mathematical Modeling, Probability, Statistics, Real Analysis, Differential Equations
Skills: Python (Pandas, NumPy, scikit-learn), Excel, Spanish (native), Data Visualization, Statistical Analysis

Quantitative Research Projects

Directed Reading Program Tallahassee, FL
Machine Learning for Auto Loans August 2024 - May 2025

- Built and evaluated logistic regression models in Python (pandas, NumPy, scikit-learn) using 25,000+ consumer lending observations to analyze loan approval outcomes and assess predictive fairness across borrower populations.
- Performed exploratory data analysis and feature engineering across quantitative and categorical borrower variables.
- Investigated bias in predictive lending models, weighing classification performance, interpretability, and fairness across borrower populations through empirical evaluation.
- Summarized statistical findings with visualizations and written analysis for technical and non-technical readers.

Time-Series Data Analysis and Predictive Modeling August 2025 - March 2026

- Studied probabilistic von Mises–Fisher mixture and regime-switching methods for high-dimensional EEG time-series data, working with 500,000+ observations in Python and Pandas.
- Evaluated temporal features including entropy, volatility, transition dynamics, and state persistence to interpret latent behavioral patterns.
- Interpreted and presented results at national conferences and academic symposiums, translating complex mathematics into clear visual and verbal explanations; used GitHub for version control.

Experience

Florida State University Tallahassee, FL
Learning Assistant - Discrete Mathematics August 2024 – May 2026

- Supported 150+ students through lectures, problem-solving sessions, and 2+ hours of weekly office hours, explaining abstract mathematical and statistical concepts to students with varying technical backgrounds.
- Provided feedback on 50+ assignments weekly with strong attention to detail and consistency.

Awards, Symposiums, and Professional Development

Presenter, **American Mathematical Society Joint Mathematics Meetings** - Washington, D.C. (January 2026)
3rd Place, Tam Family Poster Competition - Florida State University (April 2025)
Presenter, **Troy Mathfest** - Troy University (April 2025)

Leadership and Activities

Pi Mu Epsilon Tallahassee, FL
Vice President February 2024 - May 2026

- Coordinated weekly chapter operations, outreach events, invited-speaker presentations, and leadership initiatives, balancing event logistics, team communication, and competing deadlines.